Algorithm 1 Skip Module $\mathcal{S}$: Motion-Based Frame Gating

Require: Video stream $\{f_1, f_2, ...\}$, motion detector $\mathcal{D}$, scale factor α, block size B, block active threshold τ
Ensure: Skip decisions $\{s_1, s_2, ...\}$, where $s_t \in \{0, 1\}$

$\tilde{f}_{\text{prev}} \leftarrow$ null ▷ Initialize previous downsampled frame
for each incoming frame f_t **do**
  Step 1: Pre-process
  Downsample f_t by α to obtain $\tilde{f}_t$
  Let $B_s \leftarrow B \cdot \alpha$ ▷ Effective block size on downsampled frame
  Pad $\tilde{f}_t$ to be divisible by B_s
  Step 2: Compute Foreground Mask
  if $\tilde{f}_{\text{prev}} =$ null **then** ▷ No previous frame on first iteration
    $s_t \leftarrow 1$ ▷ Run inference on first frame
    $\tilde{f}_{\text{prev}} \leftarrow \tilde{f}_t$
    continue
  end if
  $\mathbf{M}_t \leftarrow \mathcal{D}(\tilde{f}_t, \tilde{f}_{\text{prev}})$ ▷ $\mathcal{D}$: e.g., FrameDiffDet or AccMotionDet
  Partition $\mathbf{M}_t$ into a grid of non-overlapping blocks of size B_s
  Mark a block as *active* if its foreground pixel ratio $> \tau$
  Step 3: Gating Decision
  if any active block exists **then**
    $s_t \leftarrow 1$ ▷ Motion detected, run DL inference
  else
    $s_t \leftarrow 0$ ▷ No motion, skip inference
  end if
  $\tilde{f}_{\text{prev}} \leftarrow \tilde{f}_t$ ▷ Update previous frame for next iteration
  yield s_t
end for